

Features and Distractors

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can choose recorded properties that preserve distinctions needed for a stated task.

I can show why identical feature records can hide examples with different target labels.

I can reject an answer copied into the inputs and test a proposed feature with a new case.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 9: Features and Distractors

A feature should help the task and be available at decision time.



THINKER MISSION

Rank book features; reject cover color as a shortcut.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Two annotators agree that a calendar card is ready. What must you still check before treating that label as correct?

This week's end goal

By the end, I can test a distracting feature with controlled changes rather than confounding clues.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use features and distractors to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number. The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality. Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

A feature is a recorded input property, such as width in centimetres. The target label is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the inputs. A card number can help us find a record, but knowing its number does not tell us its dimensions.

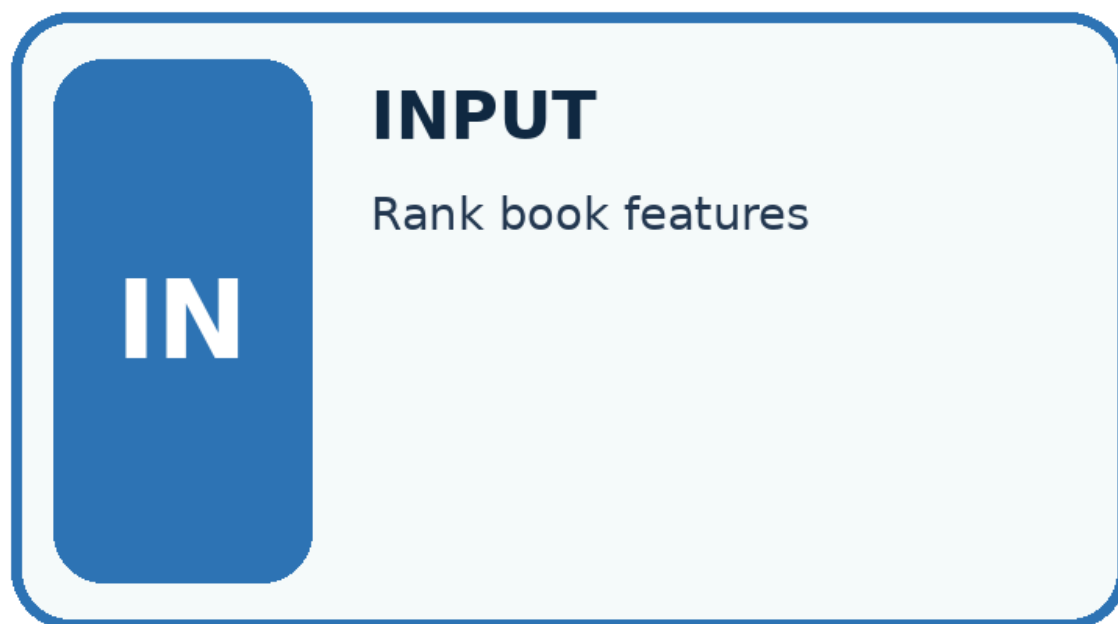
This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a learned model would need. Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

A feature should help the task and be available at decision time.





OUTPUT

reject cover color as a shortcut.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

The same cards under two feature choices

Original card · Choose the inputs · Resulting feature record

Original card: A is 10 by 7 cm and fits

Choose the inputs: Keep length only

Resulting feature record: 10 cm

Original card: B is 10 by 9 cm and is too large

Choose the inputs: Keep length only

Resulting feature record: 10 cm

Original card: A is 10 by 7 cm and fits

Choose the inputs: Keep length and width

Resulting feature record: 10 cm and 7 cm

Original card: B is 10 by 9 cm and is too large

Choose the inputs: Keep length and width

Resulting feature record: 10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same input, 10, is paired with two different correct answers. It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that input alone.

A feature record needs a consistent order and units. Here we write length in centimetres first, then width in centimetres. The record 10, 7 means something different if someone silently switches the order or records

millimetres. Write the definitions alongside the data.

Connect the picture and the explanation

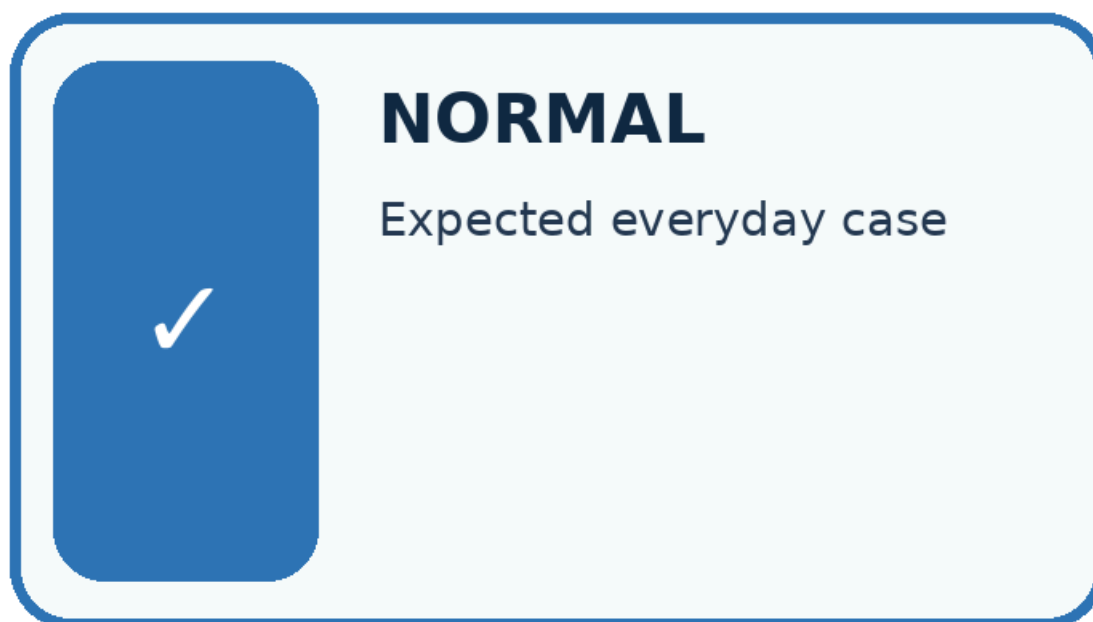
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

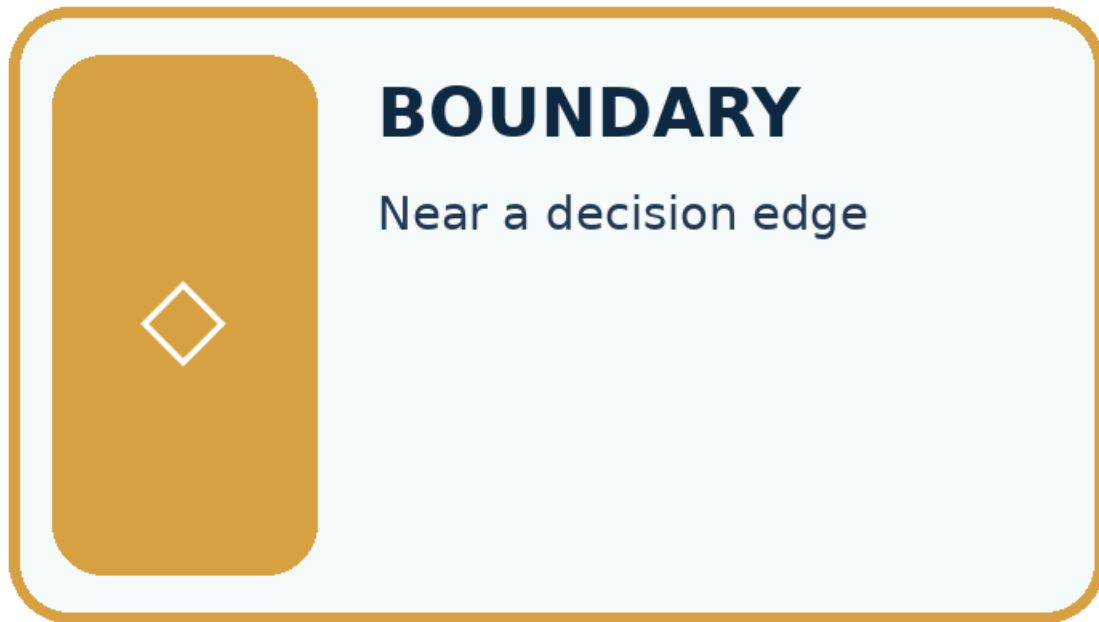
4. Compare the cases

CASE GALLERY

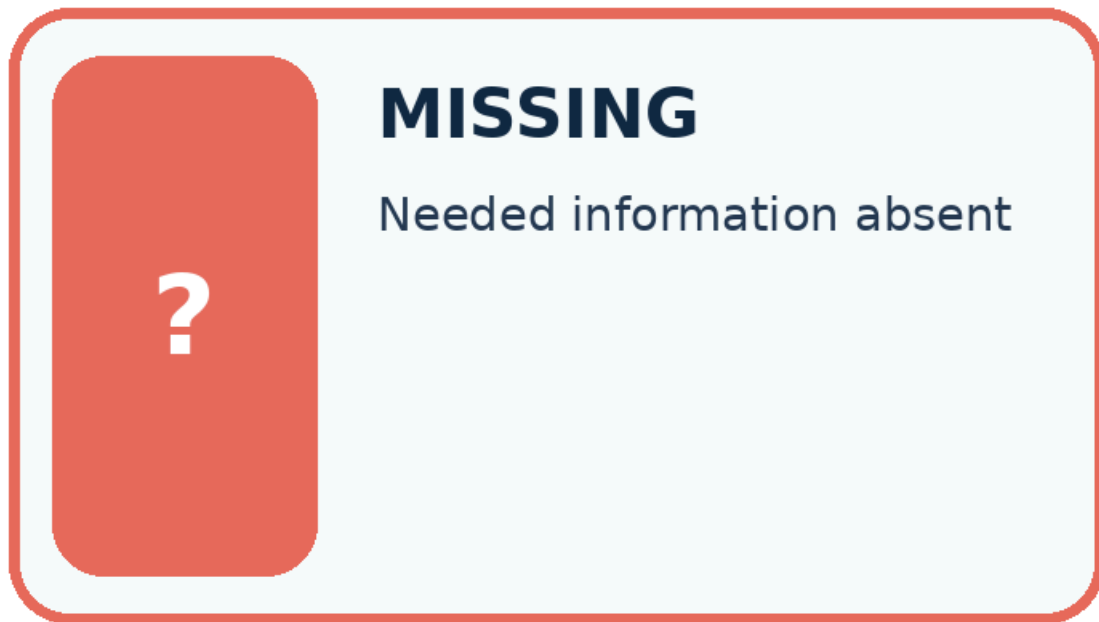
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.





MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The first controlled pair yields mystery for triangle and science for circle.

Prediction: That pair is evidence of unwanted logo sensitivity.

Reason: The subject description stayed mystery; only the irrelevant logo changed.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

FEATURE

information used to make a decision

2

RELEVANT

connected to the goal

3

DISTRACTOR

available but unhelpful
information

4

SHORTCUT

an easy clue that may fail
outside the examples

Word · My own example · Not this / boundary

Word: feature

My own example: An input attribute or representation used by a model or decision rule.

Not this / boundary: _____

Word: relevant

My own example: Useful to the particular question, target or decision being considered.

Not this / boundary: _____

Word: distractor

My own example: A detail that draws attention but does not help answer the current question.

Not this / boundary: _____

Word: shortcut

My own example: A convenient but unreliable cue used instead of the relationship the task requires.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Target label

Meaning: The answer a supervised learning task asks a model to predict

Word: Feature record

Meaning: The selected input values for one example, in a defined order

Word: Relevant feature

Meaning: An input property that helps make distinctions needed for the stated task

Word: Leakage

Meaning: Information used during development that would not legitimately be available when making the intended prediction

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 9 • WORKED EVIDENCE

- 1 Goal: recommend enjoyable readable books.
- 2 Useful: genre match, reading level, length preference.
- 3 Distractor: cover color may match past labels by accident.
- 4 Deployment check: every chosen feature must exist before recommending.

1. Goal: recommend enjoyable readable books.
2. Useful: genre match, reading level, length preference.
3. Distractor: cover color may match past labels by accident.
4. Deployment check: every chosen feature must exist before recommending.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Card measurements and colour · Label under the guide · What the comparison shows

Card measurements and colour: 10 by 7 cm and blue

Label under the guide: Fits

What the comparison shows: Compare with the next card

Card measurements and colour: 10 by 9 cm and blue

Label under the guide: Too large

What the comparison shows: Same length and colour but width changes the label

Card measurements and colour: 13 by 7 cm and blue

Label under the guide: Too large

What the comparison shows: Same width and colour as the first card but length changes the label

Card measurements and colour: 10 by 7 cm and red

Label under the guide: Fits

What the comparison shows: Changing only colour does not change this guide

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule. That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Suppose every fitting card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the shortcut. Check a proposed feature on deliberately different cases before trusting the practice score.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Features and Distractors

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

The facts we will use

Training contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test outputs have not been supplied.

Features and Distractors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Spot confounding

Topic and logo always move together in the training set.



2. Hold topic fixed

Compare the same mystery description with triangle versus circle logo.



3. Hold logo fixed

Compare mystery versus science descriptions with identical triangle logo.



4. Observe separately

Record predictions before concluding which clue the classifier uses.

CHANGE ONE THING

The first controlled pair yields mystery for triangle and science for circle.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Spot confounding

What happens and why: Topic and logo always move together in the training set.

Step: 2. Hold topic fixed

What happens and why: Compare the same mystery description with triangle versus circle logo.

Step: 3. Hold logo fixed

What happens and why: Compare mystery versus science descriptions with identical triangle logo.

Step: 4. Observe separately

What happens and why: Record predictions before concluding which clue the classifier uses.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their predictions by changing one aspect at a time. Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

What this example does not establish

Controlled observations support specific sensitivity claims; they do not reveal every internal feature a learned model uses.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Any field correlated with past choices is automatically safe to use.
Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Before adopting a feature set, state the task, name the inputs with units, and ask whether two different answers can share the same input record. Construct such a pair if you can. Then check that each input is

available at the time the answer is needed. These checks can reveal a broken representation before any model is trained.

For our known measurements and fixed orientation, length and width are sufficient to apply the paper rule exactly. That does not establish whether a real parcel will fit through a real opening: thickness, shape and measurement error were excluded. If the task changes, review the features again. More columns are not automatically better; each one needs a reason to be collected and used.

Someone claims that a model using only card colour is reliable because it matches every practice label. Give a new card that challenges this claim and explain what should be checked instead

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

After measuring a card, a teacher adds a sticker saying fits or too large. A learner proposes using sticker text as an input to predict the label before that inspection. The sticker seems excellent in the completed records because it contains the answer. But it does not exist at the time of the intended prediction. This is leakage: development has used information the real task cannot provide.

Draw a line at the moment the system must answer. On the before side, put the permitted measurements available then. On the after side, put the final label sticker. Keep labels available for teaching and checking predictions, but do not feed the sticker back in as an input for this task.

A card is 12 cm long and its width is unreadable. Someone proposes using the label sticker that will be added after inspection. Which information is usable before inspection? Explain why it does not establish the label

Practise with less help: Prove that one measurement is insufficient

Use workbook W2 for the connected example “Features and useful distinctions”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case.

Each person writes their own explanation.

Check your reasoning

Use the worked case for Features and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Rank eight candidate features by relevance, availability, privacy, and shortcut risk. Defend the top three and reject two tempting distractors.

Rank eight candidate features by relevance, availability, privacy, and shortcut risk. Defend the top three and reject two tempting distractors.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

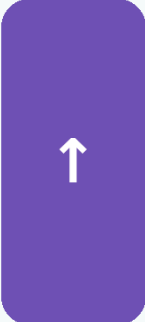
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER
Use explainable traits that connect to the reader's stated preferences.

THIS WEEK'S CHECKPOINT Use explainable traits that connect to the reader's stated preferences.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Test recommender features for shortcuts, keeping relevant subject information separate from superficial catalogue styling.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Rank book features; reject cover color as a shortcut.

CHALLENGE Rank book features; reject cover color as a shortcut.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

The question

Choose useful evidence and possible distractions.

Worked explanation

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental shortcuts. Test on new books.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Useful evidence versus a shortcut

Evidence or stage · Value or result

Evidence or stage: Target

Value or result: Contains animal characters

Evidence or stage: Animal tag

Value or result: Relevant if accurate

Evidence or stage: Cover colour

Value or result: Possible distraction

Evidence or stage: Shelf number

Value or result: Possible accidental shortcut

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful evidence and possible distractions.

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental shortcuts.
Test on new books.

14. Try a new case independently

Independent case: Changed independent case

Training covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Predict whether a book is short using page count and cover colour. Which feature directly supports the defined label? Give a counterexample to colour.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use explainable traits that connect to the reader's stated preferences.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

All animal books in training have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.